

Ex. No.: 6c)
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PRIORITY SCHEDULING

Aim:

To implement priority scheduling technique

Algorithm:

1. Get the number of processes from the user.
2. Read the process name, burst time and priority of process.
3. Sort based on burst time of all processes in ascending order based priority 4.
4. Calculate the total waiting time and total turnaround time for each process 5.
5. Display the process name & burst time for each process.
6. Display the total waiting time, average waiting time, turnaround time

Program Code:

```
#include <stdio.h>
```

```
int main () {
```

```
    int n, i, j, temp;
```

```
    int bt[20], p[20], pr[20], wt[20], tat[20], ct[20];
```

```
    printf("Enter the number of processes: ");
```

```
    scanf ("%d", &n);
```

```
    for (i=0; i<n; i++) {
```

```
        printf("Enter Burst Time and Priority for process
```

```
                %d: ", i+1);
```

```
        scanf ("%d %d", &bt[i], &pr[i]);
```

```
        pr[i] = i+1;
```

```
    }
```

```

for (i=0; i<n-1; i++) {
    for (j=i+1; j<n; j++) {
        if (pr[i] > pr[j]) {
            temp = pr[i];
            pr[i] = pr[j];
            pr[j] = temp;
        }
        temp = bt[i];
        bt[i] = bt[j];
        bt[j] = temp;
    }
    temp = p[i];
    p[i] = p[j];
    p[j] = temp;
}

```

priority

burst time

process id.

```

}
}
int current_time = 0;
for (i=0; i<n; i++) {
    current_time += bt[i];
    ct[i] = current_time;
}
for (i=0; i<n; i++) {
    tat[i] = ct[i];
}
for (i=0; i<n; i++) {
    wt[i] = tat[i] - bt[i];
}
printf ("Process\t BT\t Priority\t CT\tTAT\t WT\n");
for (i=0; i<n; i++) {
    printf ("P%d\t %d\t %d\t %d\t %d\t %d\n",
            p[i], bt[i], pr[i], ct[i], tat[i], wt[i]);
}
return 0;
}

```

42

Input:

Enter the number of processes: 4

Enter the Burst time and Priority for Process 1: 6 2

Enter the Burst time and Priority for Process 2: 8 1

Enter the Burst time and Priority for Process 3: 7 3

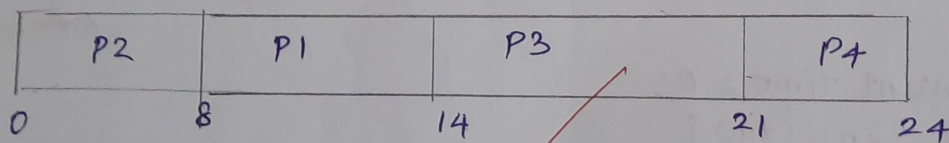
Enter the Burst time and priority for Process 4: 3 4

Output:

Process	BT	Priority	CT	TAT	WT
P2	8	1	8	8	0
P1	6	2	14	14	8
P3	7	3	21	21	14
P4	3	4	24	24	11

Process

Gantt chart:



Sample Output:

```
C:\Users\admin\Desktop\Untitled1.exe
Enter Total Number of Process:4
Enter Burst Time and Priority
P[1]
Burst Time:6
Priority:3
P[2]
Burst Time:2
Priority:2
P[3]
Burst Time:14
Priority:1
P[4]
Burst Time:6
Priority:4
Process    Burst Time    Waiting Time    Turnaround Time
P[3]       14             0               14
P[2]       2             14              16
P[1]       6             16              22
P[4]       6             22              28
Average Waiting Time-13
Average Turnaround Time-20
```

Result:

Thw the code to implement the priority scheduling has be executed successfully

[Signature]